



GENERAL PURPOSE RELAY

AUTOMOBILE RELAY

INDUSTRIAL CONTROL

MINIATURE POWER

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POWER RELAYS



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level they could communicate and move around independently due to electrostatic forces.

Realising the vision of claytronics through self-assembly of millions of catoms into synthetic reality would have a profound effect on users of electronic information. The undergoing research combines modular robotic systems, nanotechnology and computer science to create dynamic, 3D display of electronic information.

Initial research is focused on creating the basic modular building blocks of claytronics, on designing and in writing robust and reliable software programs that will shape ensembles of millions of catoms into dynamic 3D forms. The aim is to give tangible, interactive forms to information so that a user's senses can experience digital environments as if they are indistinguishable from reality. This technology will help to drive breathtaking advances in design and engineering of computing and hardware systems.

### Programmable clay

Researchers are designing a programmable clay that could morph into a working 3D replica of any person or object, based on information transmitted from anywhere in the world. The clay would be made of millions of tiny microprocessors called catoms, each less than a millimeter.

Claytronics would make possible a radical vision for the future of long-distance meetings. For instance, it may be possible to fax a copy of a speaker's body, mimicking his moves in real time and speaking in his voice, to someone sitting miles away. The project—at the moment still a long way from realisation—aims to create nanoscale robotic mechanisms with computing abilities, capable of changing form and joining together to form large-scale mechanisms or objects. With claytronics, millions of tiny individual devices would assemble into macro-scale objects, connecting and disconnecting as they move.

The current large proof-of-concept catoms of 4.4 cm connect and move via electromagnetic or electrostatic connections, much like the 'replicating' robots operating at scales where it may be difficult to control with the same technology. The catoms could have LCD or LED surfaces to produce a faintly glowing image of a person or object made of millions of tiny microbots that would actually look like the person or the object.

### Areas that would benefit

Researchers say they will have a hardware prototype of sub-millimeter electrostatic modules in five years and will be able to fax complex 3D models—of anything, from engagement rings to sports cars—by 2017. If it works, claytronics could transform communication, entertainment, medicine and more as it promises to:

1. Help users to carry around a lump of claytronics in their pockets that can reshape into any object and even act like 3D TV and create synthetic reality.
  2. This technology would enable engineers to work remotely in physically hostile environments or surgeons to perform intricate surgery on enlarged claytronic replicas of organs, while the actual organs are being worked upon by a claytronic replica of the surgeon.
  3. It may help scientists learn how to efficiently manage networks of millions of computers.
  4. It will also advance our understanding of nanotechnology.
- Many challenges need to be resolved before this vision can become reality. The hardware aspects of the task like making crude, clumsy and tiny devices work is difficult but manageable. The software challenges are, however, more staggering. Coordinating the work of a few dozen robots or programming millions to work together is incredibly difficult. ●

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